



INSTITUT
POLYTECHNIQUE
DE PARIS

IMA 4103 : COMPUTER VISION

**NEURAL-NETWORK FOR
COMPUTER VISION**

MDC. DECKY ASPANDI



OUTLINES

- Models explainability
 - Filter visualisations
 - Gradcam visualisations
- Example cases :
 - Filter visualisations
 - Filter activation and Gradcam visualisations

SECTION - 1

Models explainability Discovering the learned features

EXPLAINABILITY IN DEEP LEARNING

PREFACE

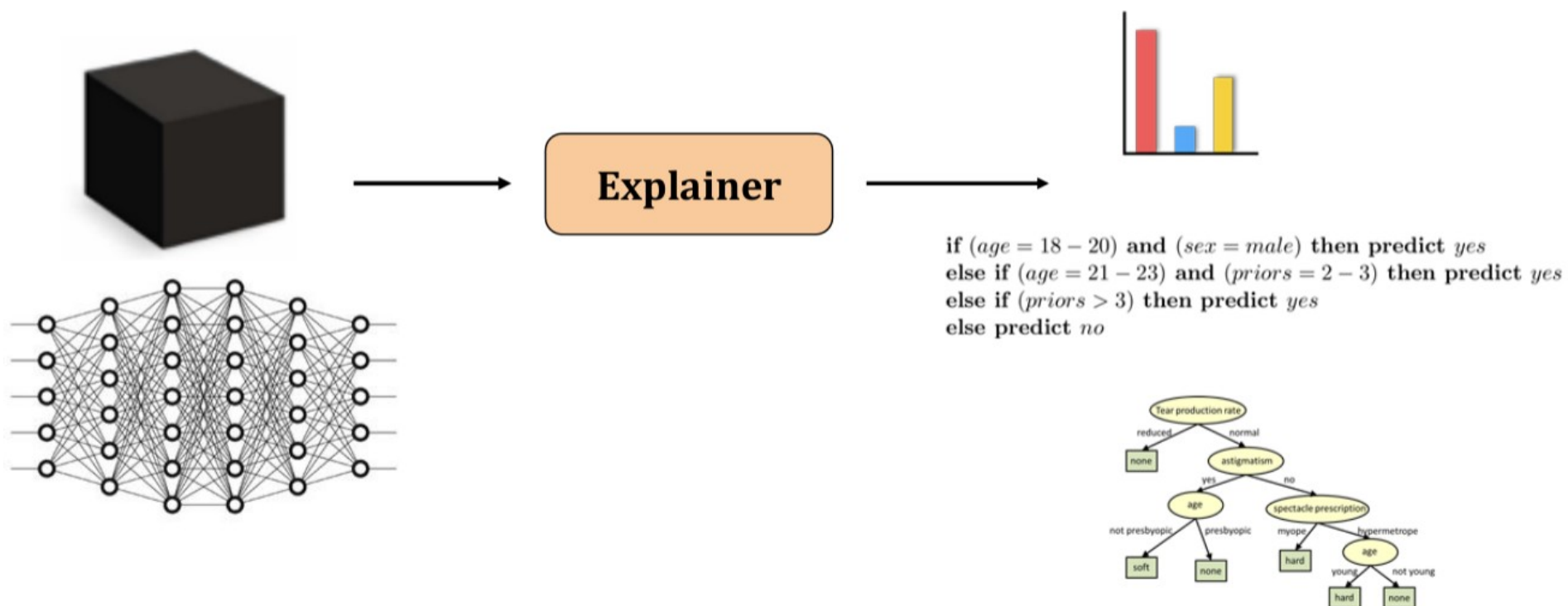
- The need for interpretability :
 - Help building trust:
 - Humans are reluctant to use ML for critical tasks
 - Fear of unknown when people confront new technologies
 - 2. Promote safety:
 - Explain model's representation (i.e. important feature)
 - providing opportunities to remedy the situation
 - 3. Allow for contestability:
 - Black-box models don't decompose the decision into sub-
 - models or illustrate a chain of reasoning



EXPLAINABILITY IN DEEP LEARNING

POSTHOC EXPLANATIONS

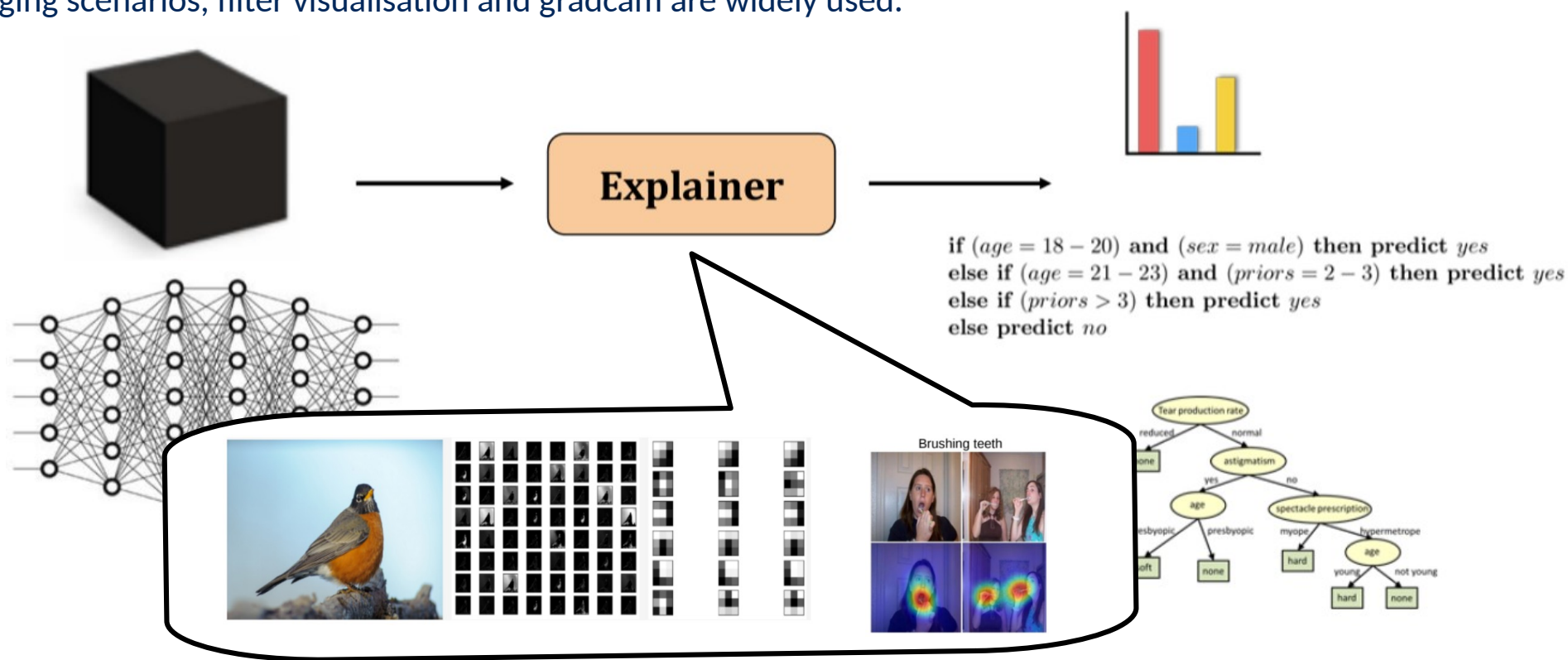
- To explain the methods after they are trained.



EXPLAINABILITY IN DEEP LEARNING

POSTHOC EXPLANATIONS

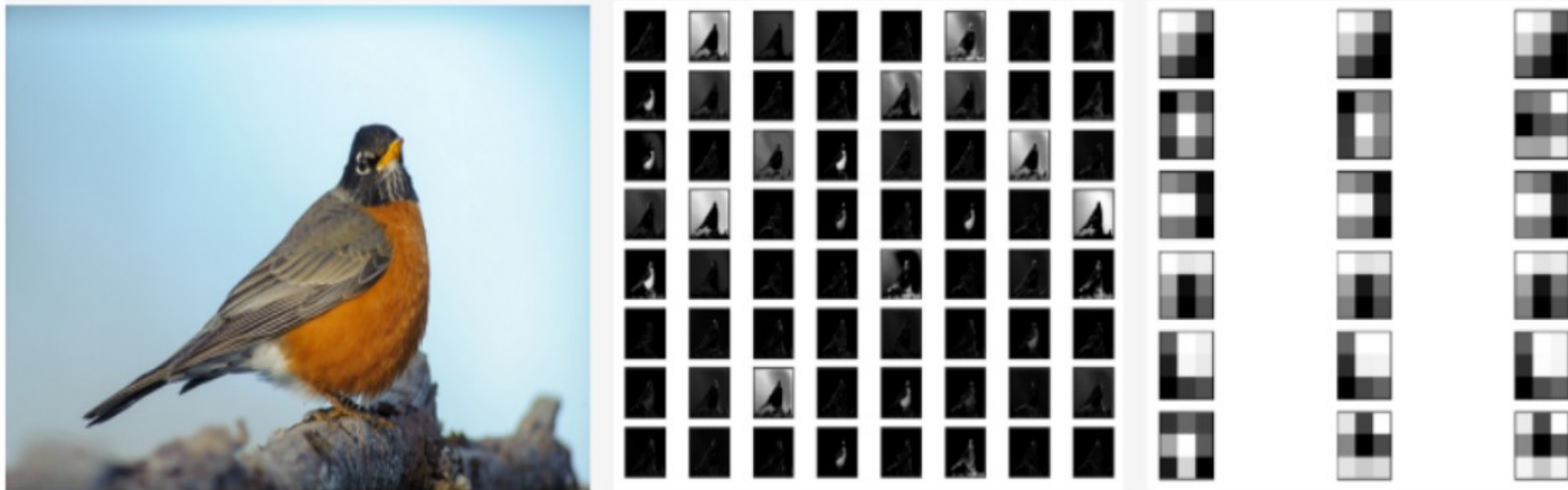
- To explain the methods after they are trained.
- In imaging scenarios, filter visualisation and gradcam are widely used.



FILTER VISUALISATION

VISUALISING INNER FILTERS

- The first things to try are:
 - Visualize the filters themselves.
 - Visualize the result of applying a learned filter to an image;



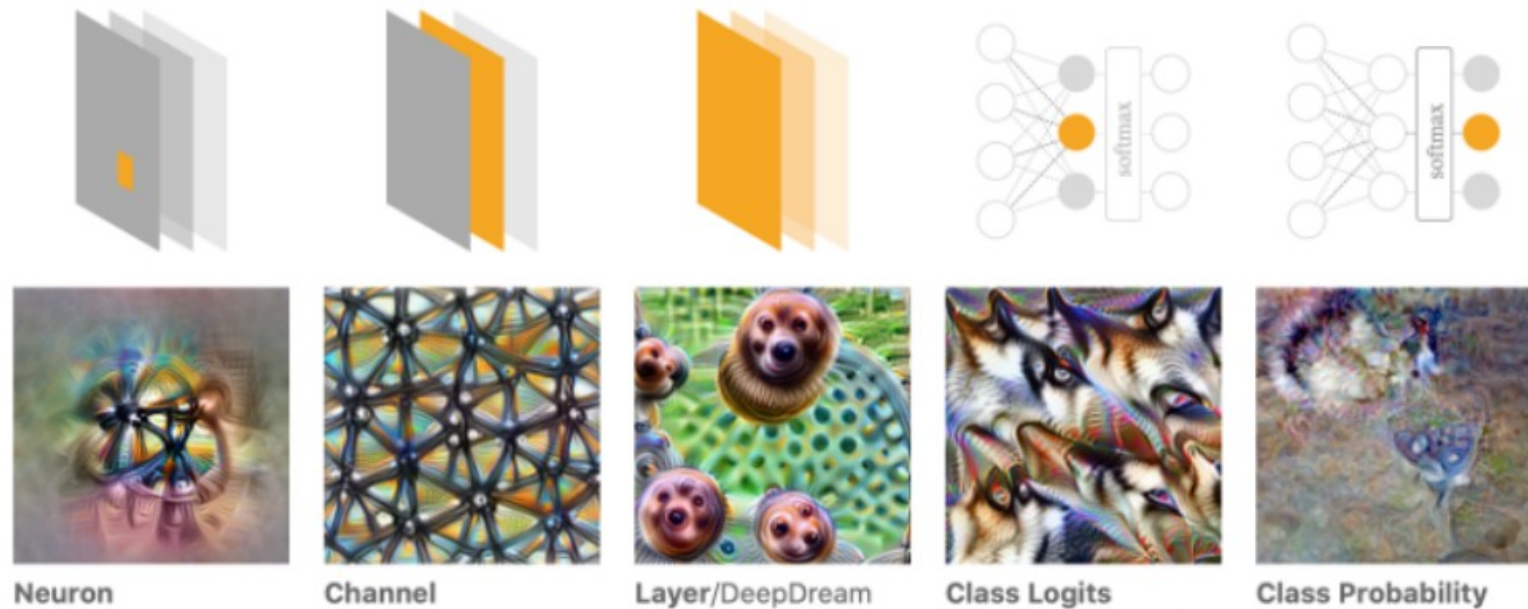
FILTER VISUALISATION

ACTIVATION MAXIMIZATION

- Rather than visualizing a particular filter or the representation of a particular image at a hidden layer, we can visualize the image that maximize the output and hence the impact of that specific filter or layer.
- That is we find image x^* that maximizes activation of a filter while holding the network weights fixed (link).

Different **optimization objectives** show what different parts of a network are looking for.

n layer index
x, y spatial position
z channel index
k class index



GRAD-CAM

WHERE THE MODEL FOCUS

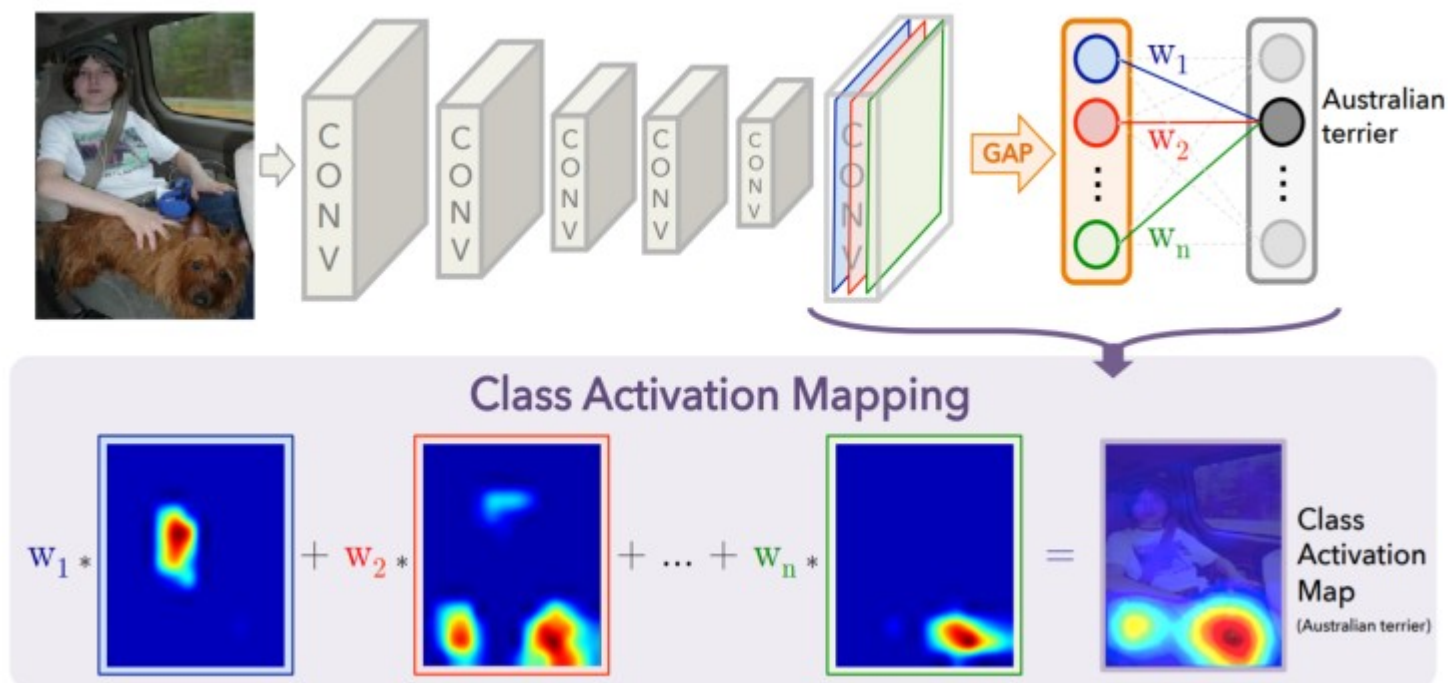
- Class Activation Maximization is another explanation method for interpreting convolutional neural networks (CNNs) introduced by Zhou et al. 2016.
- They proposed a network where the fully connected layers at the very end of the model has been replaced by a layer named Global Average Pooling (GAP) and combined with a class activation mapping (CAM) technique.



GRAD-CAM

PREFACE

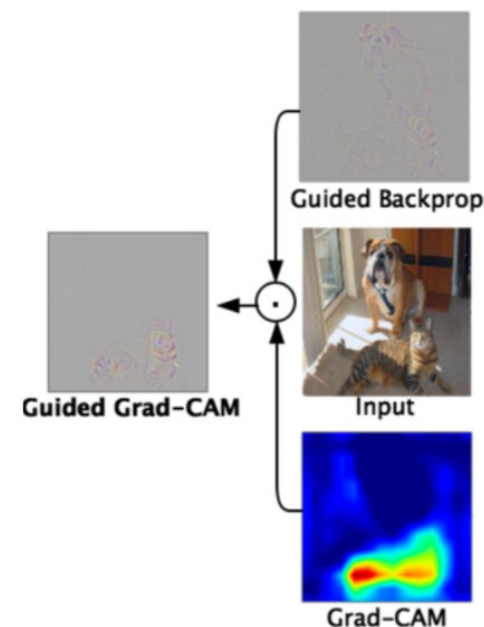
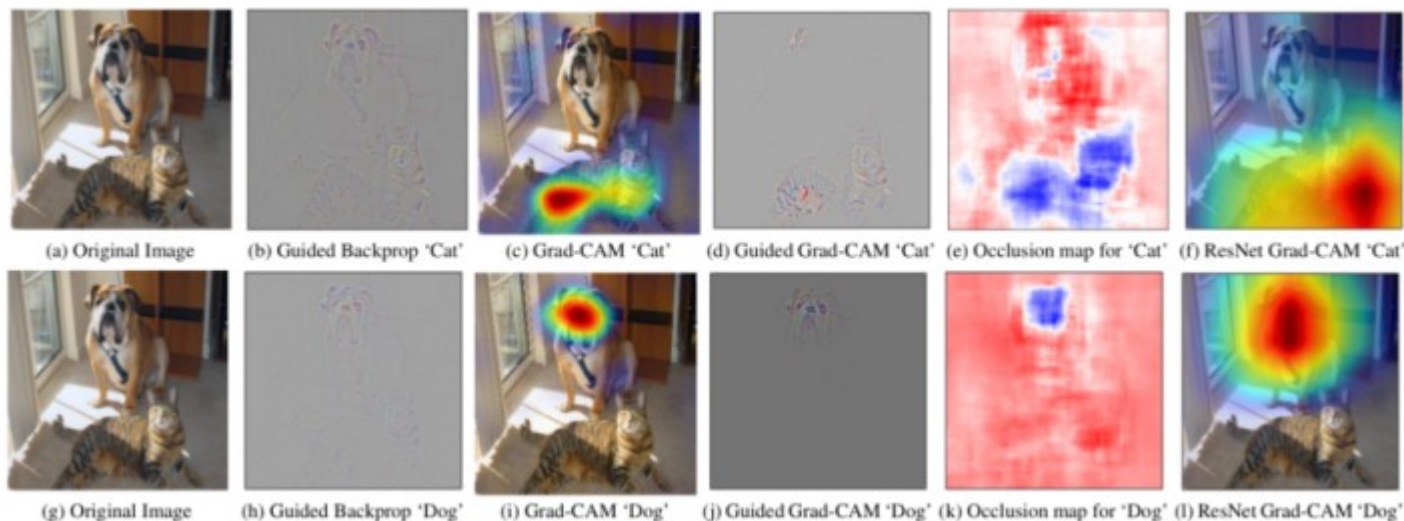
- The GAP averages the activations of each feature map and concatenates and outputs them as a vector.
- Then, a weighted sum of the resulted vector is fed to the final softmax loss layer.



GRAD-CAM

PREFACE

- As Grad-CAM can only produce coarse-grained visualizations, the authors have also combined guided-backpropagation with their method and propose Guided Grad-CAM.
- They simply do that by an element-wise multiplication of guided-backpropagation visualization as shown in the figure.



ANY QUESTIONS ?



SECTION - 2

Coding Activities : TP 3 Filter visualisation

THANK YOU

Decky Aspandi

E-mail : decky.aspandi_latif@telecom-sudparis.eu

Phone : (+33) 01 60 76 46 40

Office : Dept. ARTEMIS, Room A201.01

Address : TELECOM SudParis - 9 rue Charles Fourier -
91011 EVRY-COURCOURONNES Cedex.

